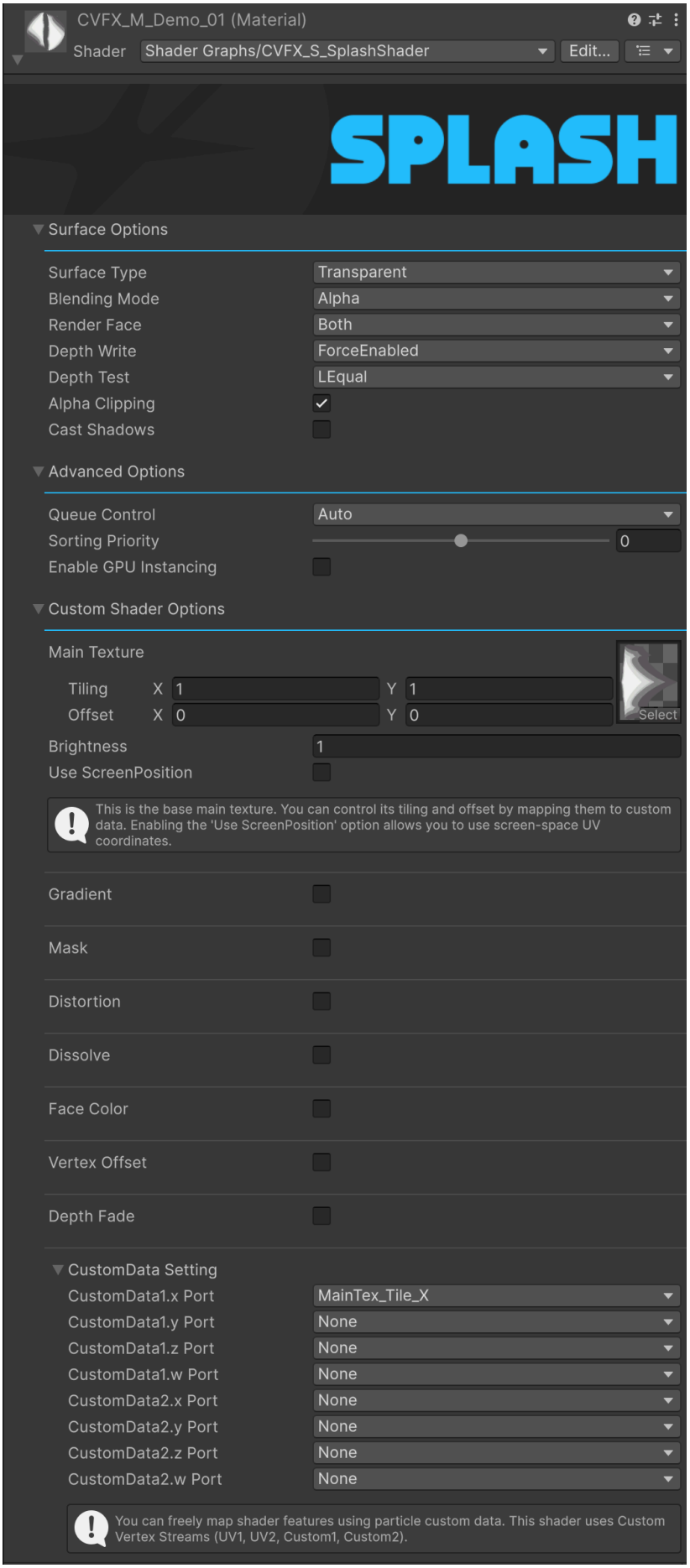
[Quick Start Guide: Splash Shader]


CVFX_M_Demo_01 (Material)
Shader
Shader Graphs/CVFX_S_SplashShader
Edit...
SPLASH
Surface Options
Surface Type
Transparent
Blending Mode
Alpha
Render Face
Both
Depth Write
ForceEnabled
Depth Test
LEqual
Alpha Clipping
Cast Shadows
Advanced Options
Queue Control
Auto
Sorting Priority
0
Enable GPU Instancing
Custom Shader Options
Main Texture
Tiling X 1 Y 1
Offset X 0 Y 0
Select
Brightness
1
Use ScreenPosition
This is the base main texture. You can control its tiling and offset by mapping them to custom data. Enabling the 'Use ScreenPosition' option allows you to use screen-space UV coordinates.
Gradient
Mask
Distortion
Dissolve
Face Color
Vertex Offset
Depth Fade
CustomData Setting
CustomData1.x Port
MainTex_Tile_X
CustomData1.y Port
None
CustomData1.z Port
None
CustomData1.w Port
None
CustomData2.x Port
None
CustomData2.y Port
None
CustomData2.z Port
None
CustomData2.w Port
None
You can freely map shader features using particle custom data. This shader uses Custom Vertex Streams (UV1, UV2, Custom1, Custom2).

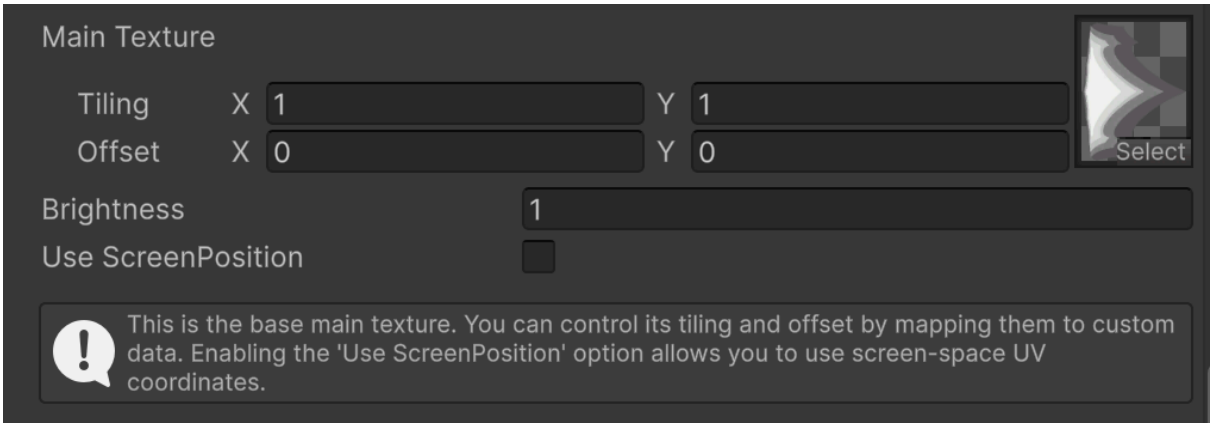
Surface Option

Same as the default shader surface options provided by the engine.

Advanced Options

Same as the default shader advanced options provided by the engine.

Custom Shader Options

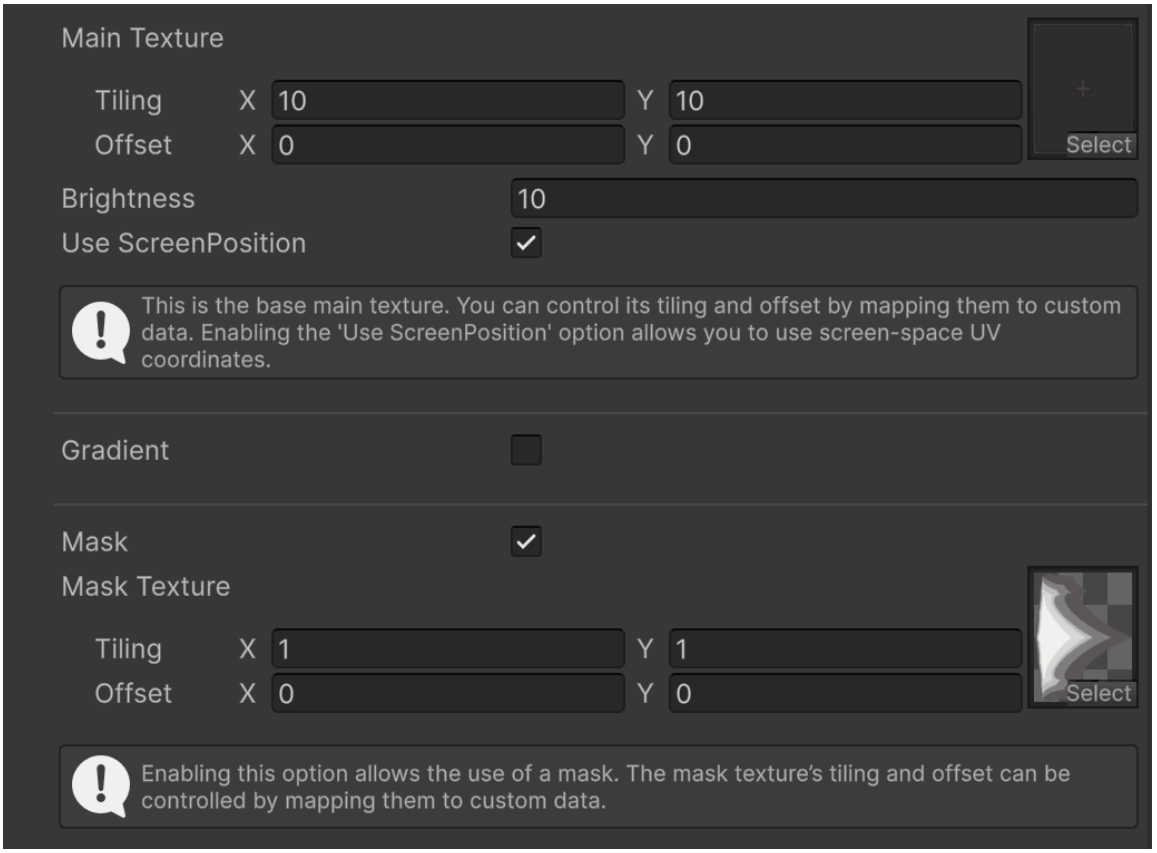


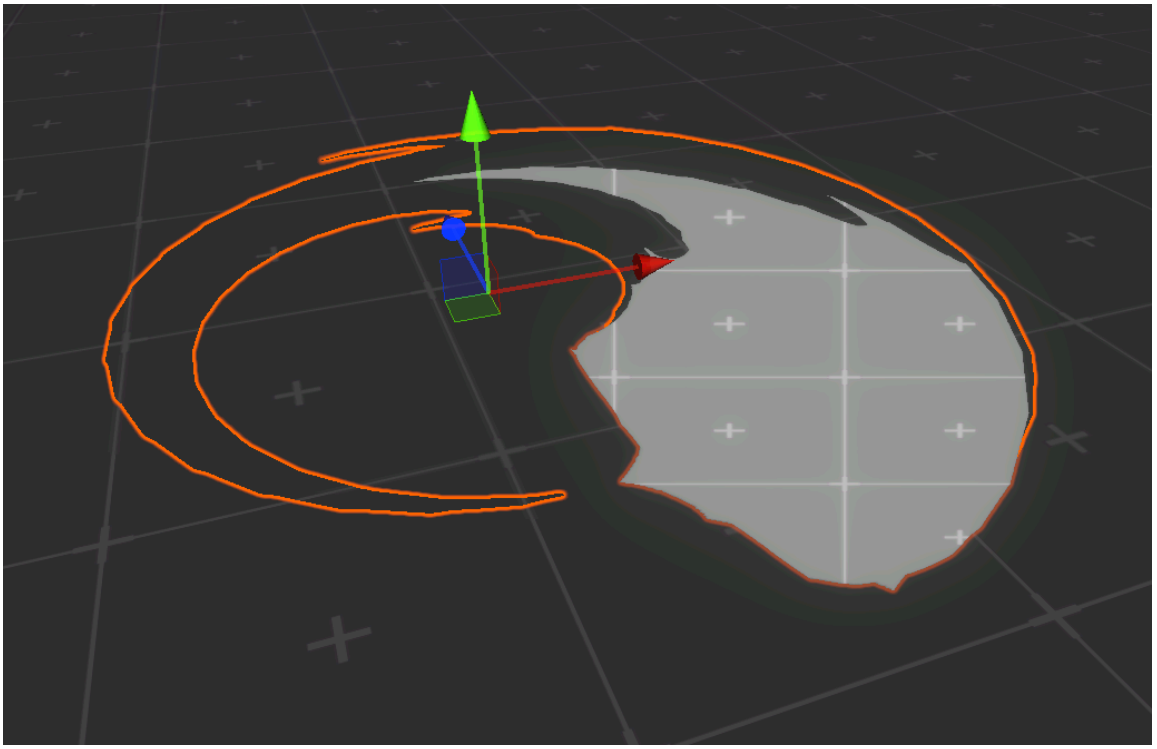
`Main Texture` Sets the main texture. Tiling and offset can be linked to custom data

`Brightness` Adjusts the brightness of the main texture.

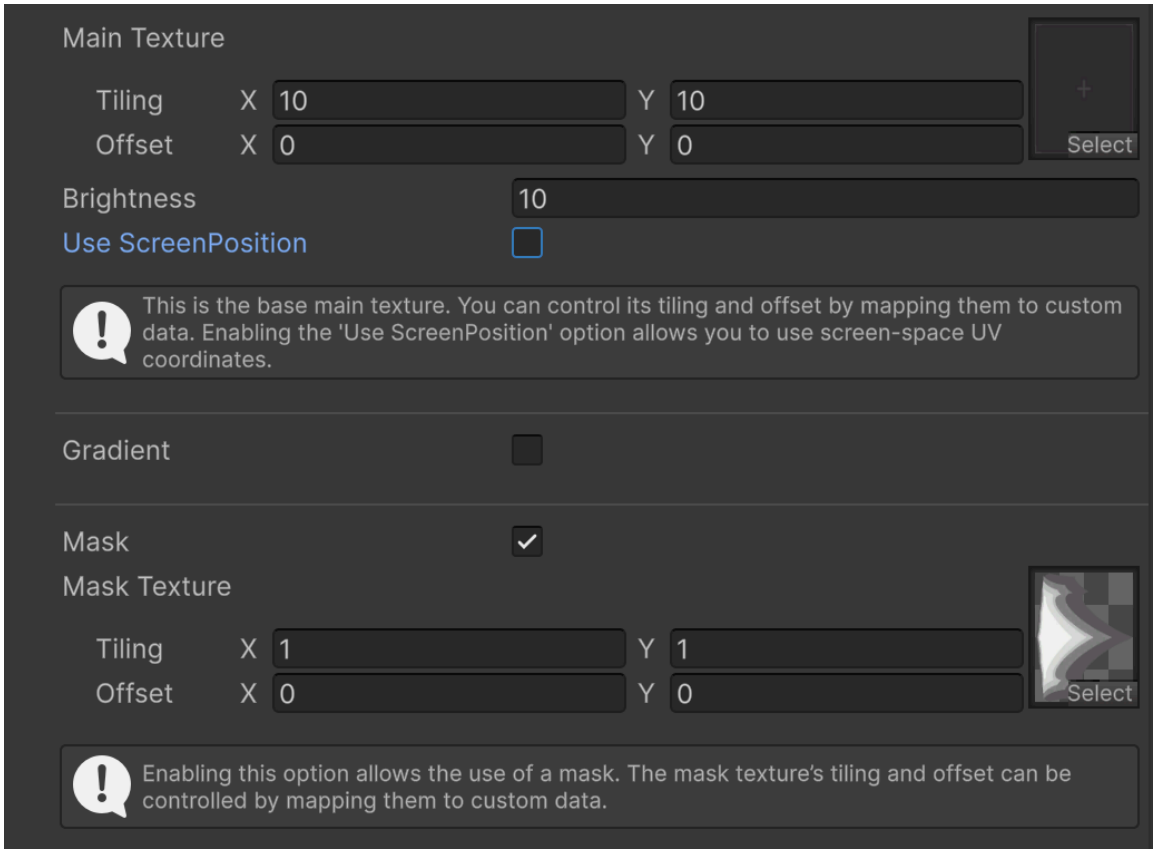
`Use ScreenPosition` Enabling this toggle uses screen position coordinates for the main texture's UVs.

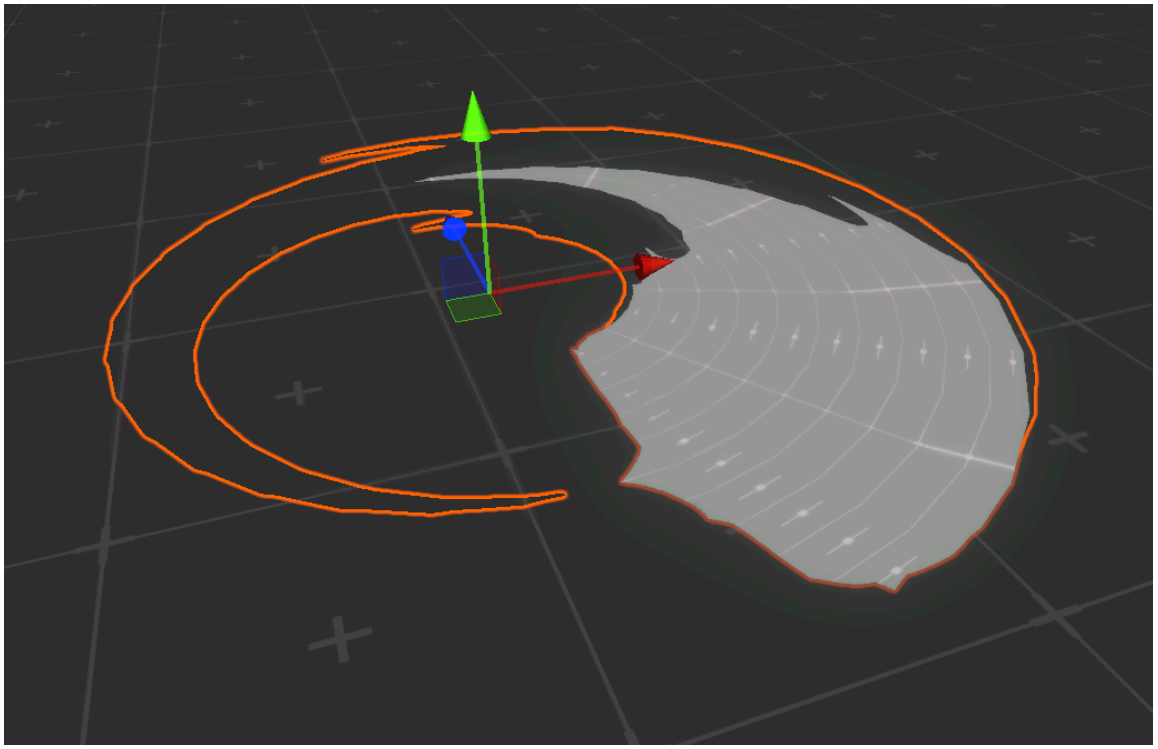
▼ Example





Uses screen-space UVs when enabled



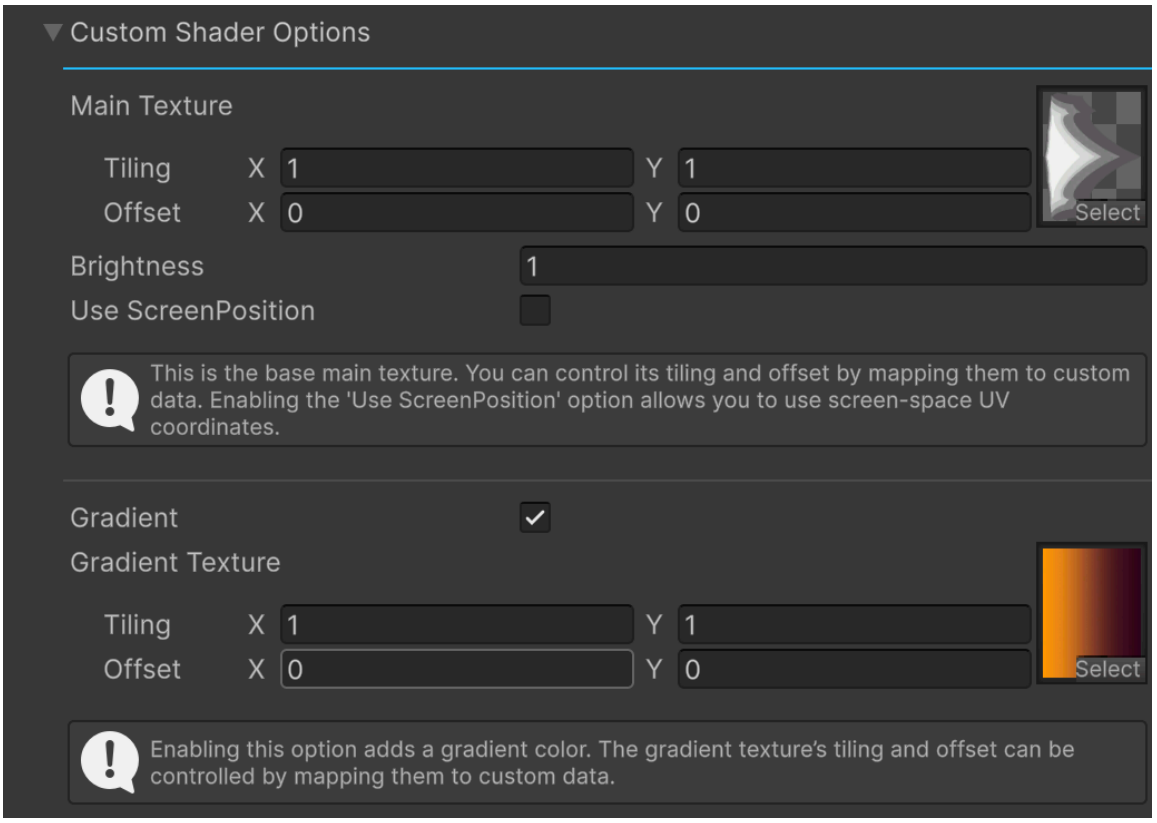


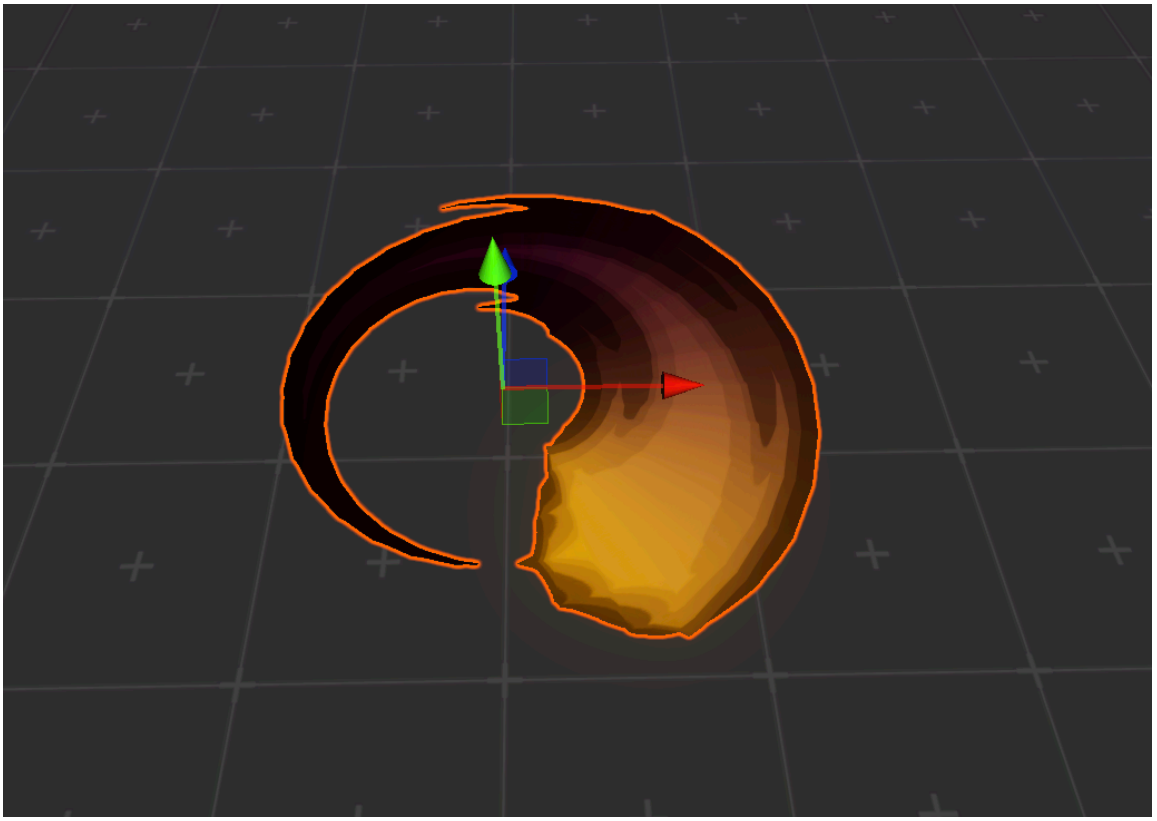
Uses mesh UVs when disabled

Enabling the Gradient toggle reveals settings related to the gradient texture. This feature multiplies the texture color based on the UV direction within the engine.

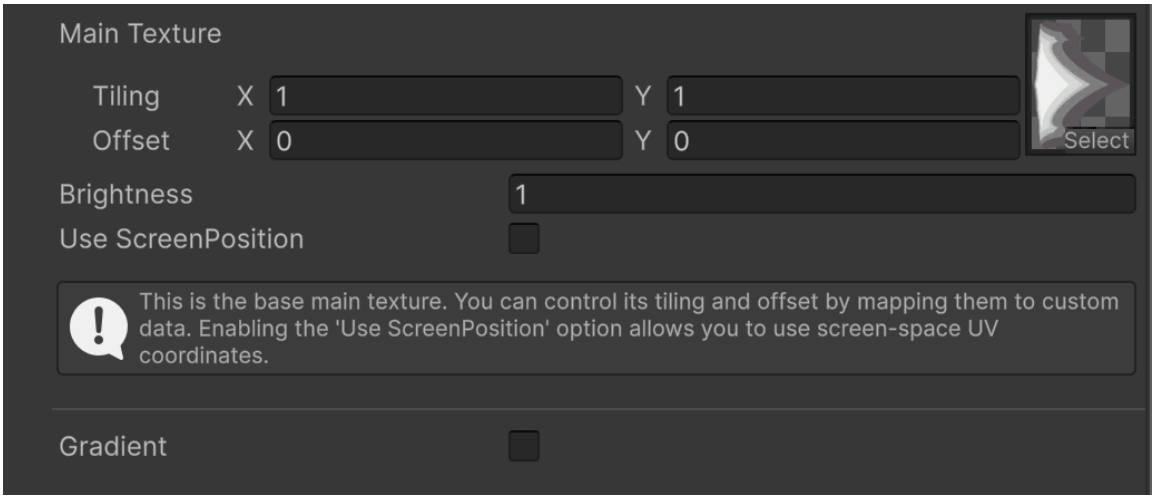
Gradient Texture	Sets the gradient texture. Tiling and offset can be linked to custom data.

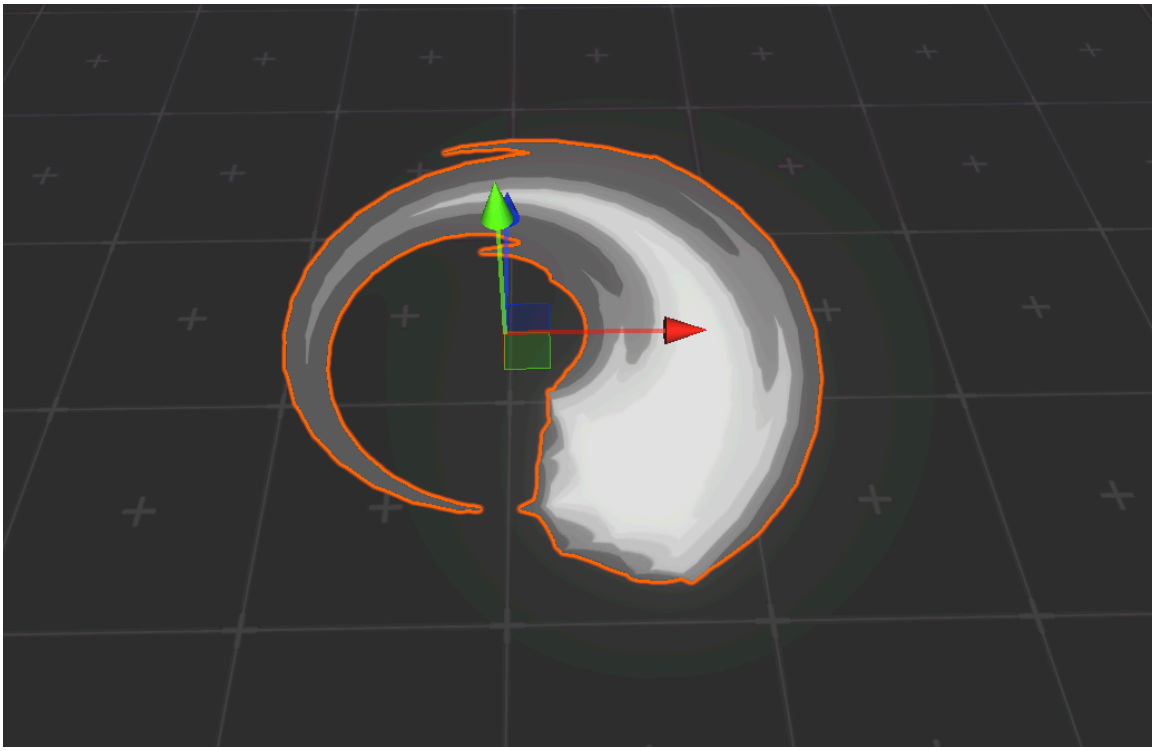
▼ Example



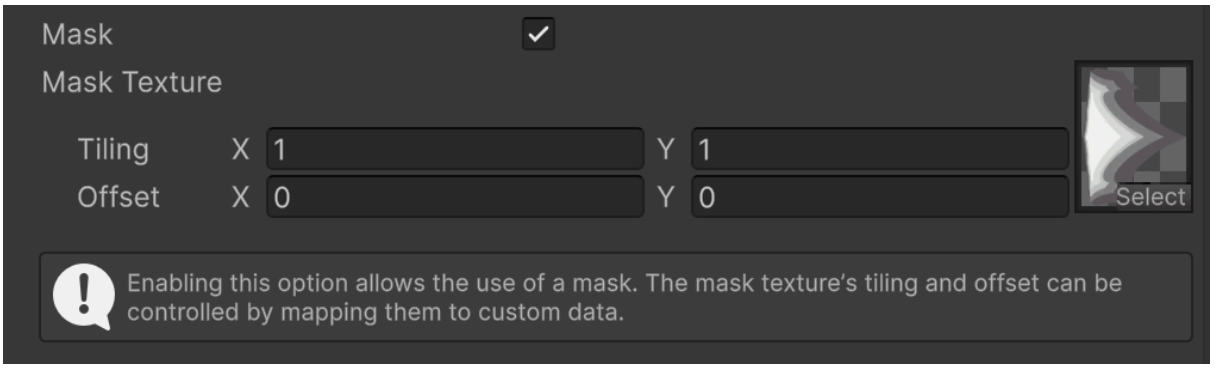


When using the gradient texture





When not using the gradient texture



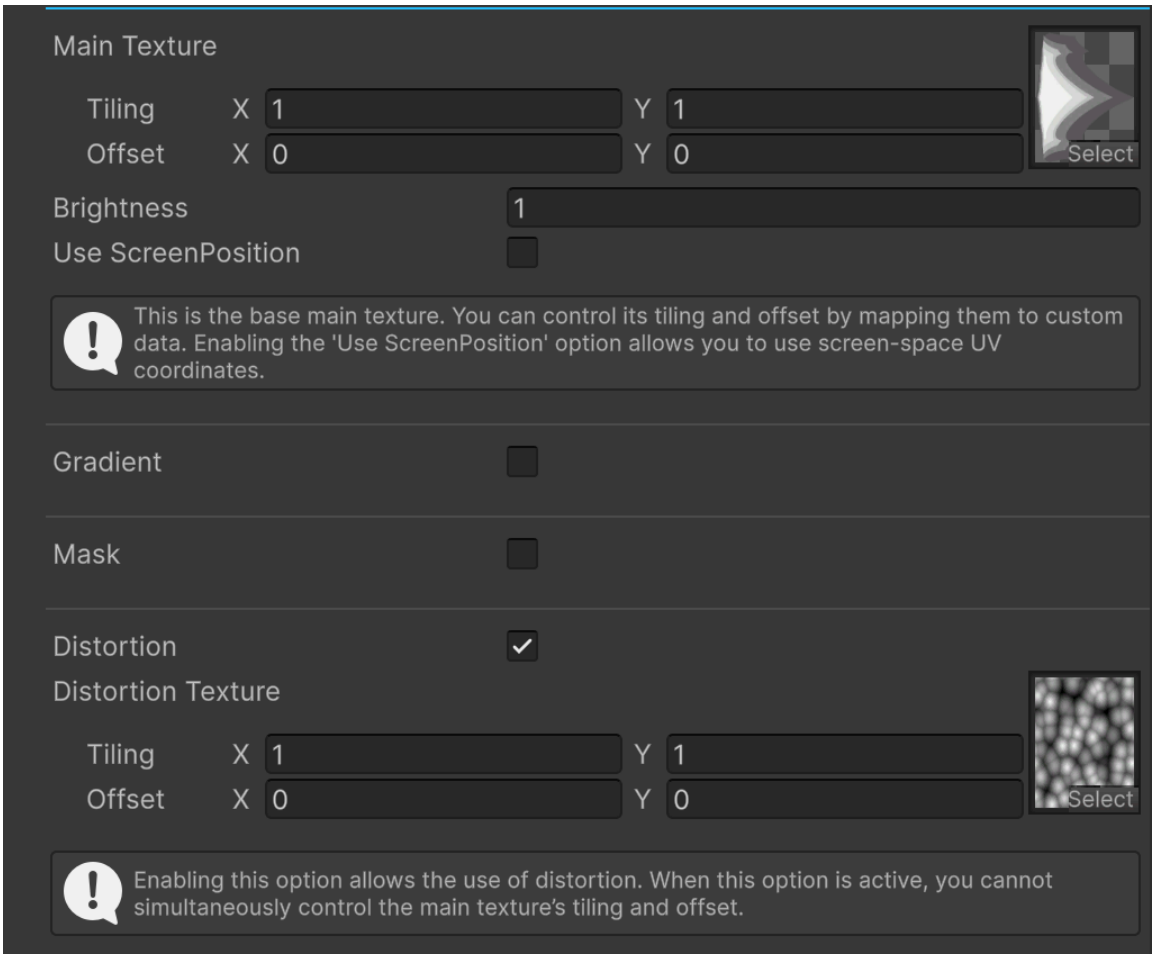
Enabling the Mask toggle reveals settings related to the mask.

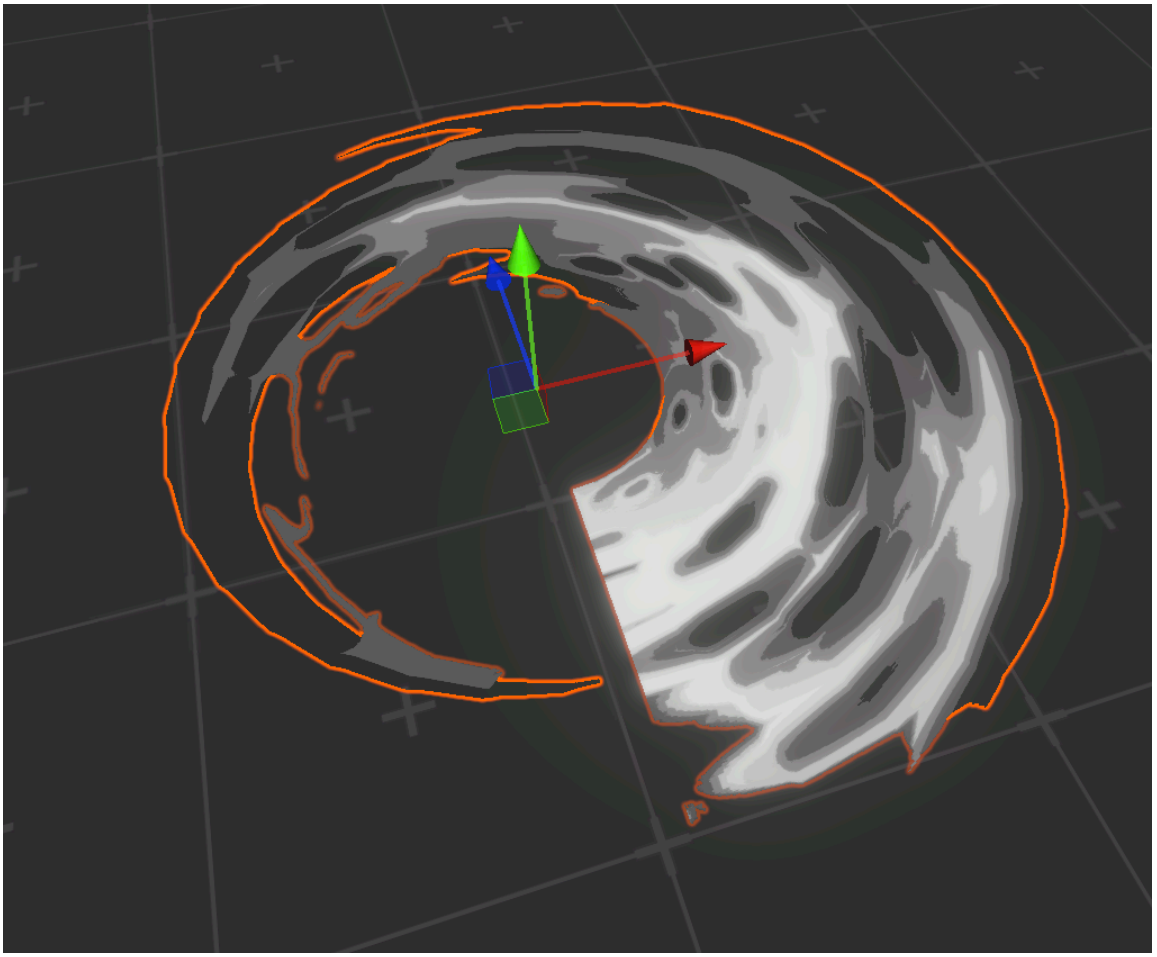
Mask Texture	Sets the mask texture. Tiling and offset can be linked to custom data.

Enabling the Distortion toggle reveals settings related to distortion.

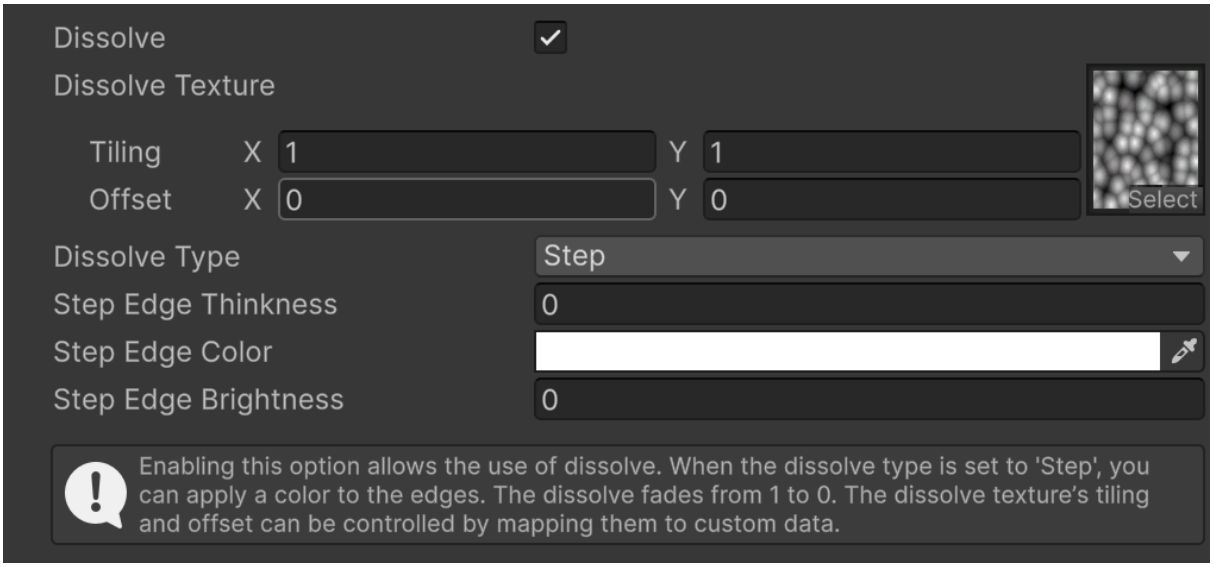
Distortion Texture	Sets the distortion texture. Tiling and offset can be linked to custom data.

▼ Example





When Distortion is applied

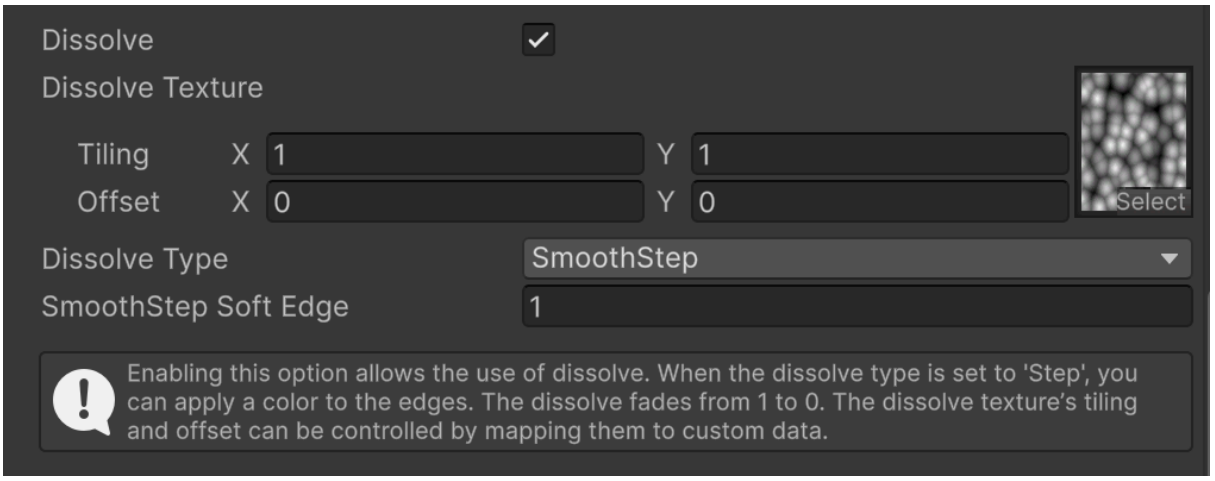


Enabling the Dissolve toggle reveals settings related to dissolve.
The dissolve value starts at 1, and as it approaches 0, the alpha fades out according to the shape of the dissolve texture.

Dissolve Texture	Sets the dissolve texture. Tiling and offset can be linked to custom data.
Dissolve Type	There are two modes: Step and SmoothStep. In Step mode, additional settings such as Edge Thickness, Color, and Brightness are displayed.

When Step mode is active, the following properties appear in the inspector

Step Edge Thinkness	Sets the edge thickness. Depending on the scene scale, a value around 0.05 is generally recommended.
Step Edge Color	Sets the edge color.
Step Edge Brightness	Sets the edge brightness.

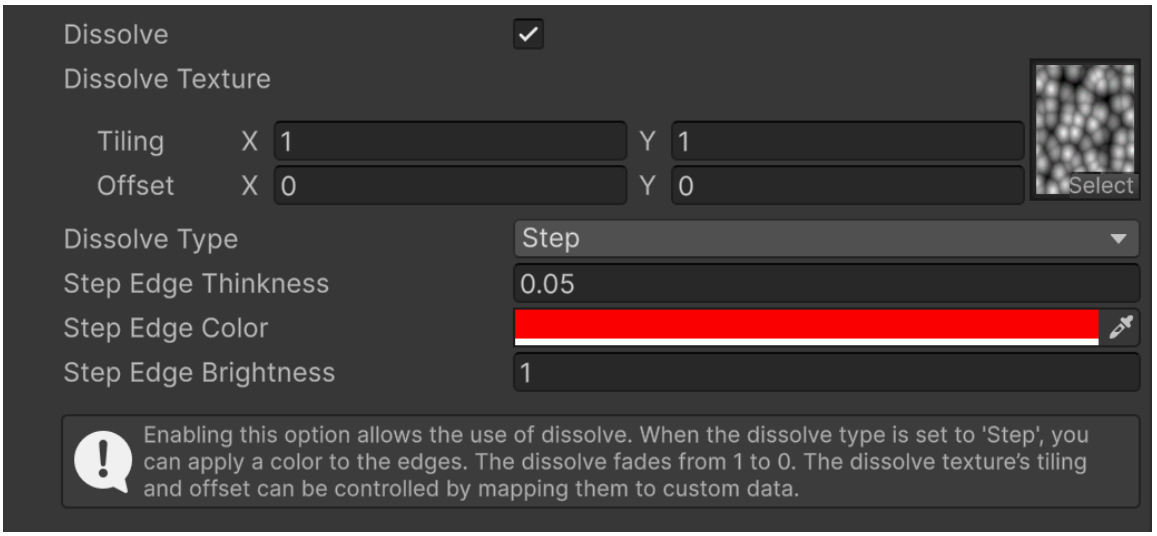


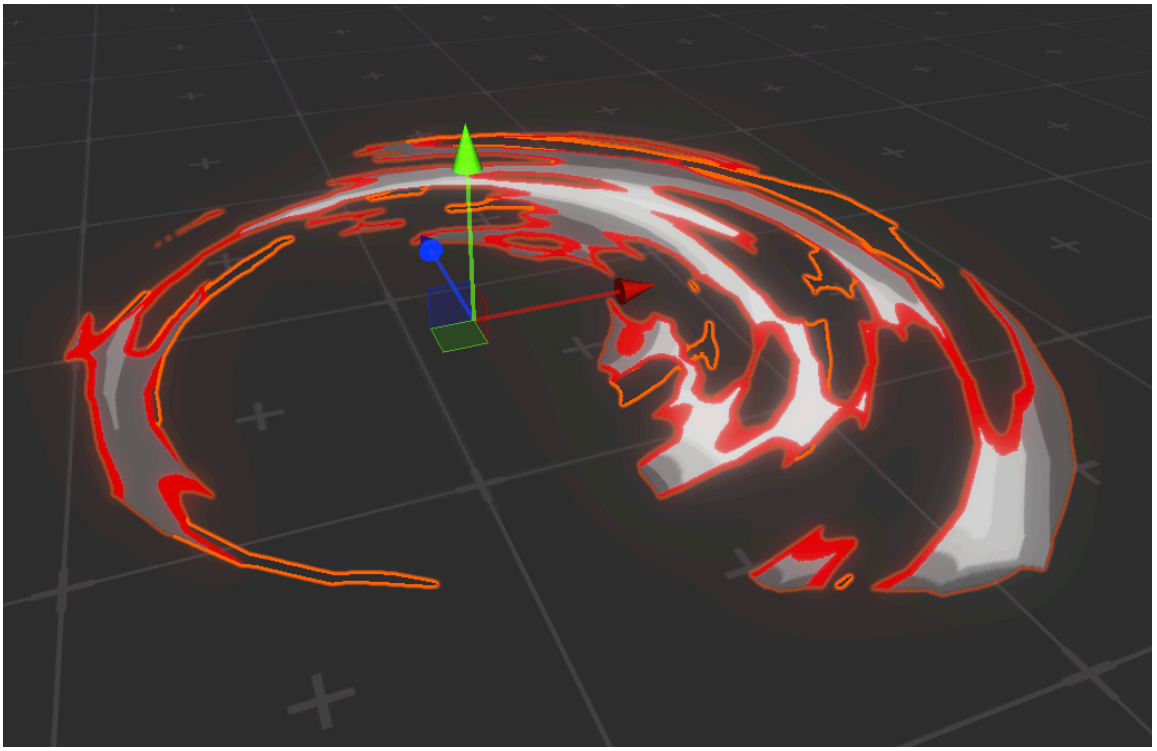
In SmoothStep mode, a setting for Soft Edge intensity is shown.

SmoothStep Soft Edge

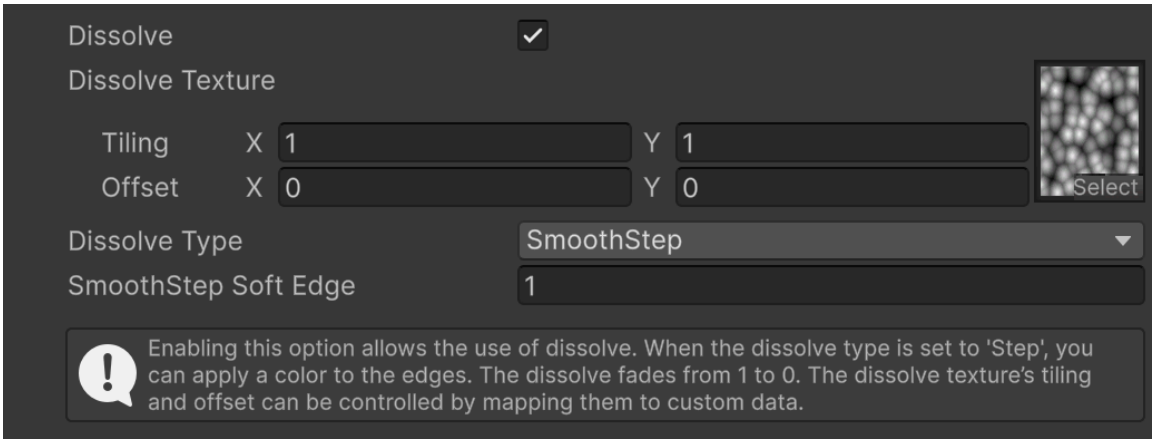
The default value is 1. As it approaches 0, the edge becomes sharper, similar to Step mode.

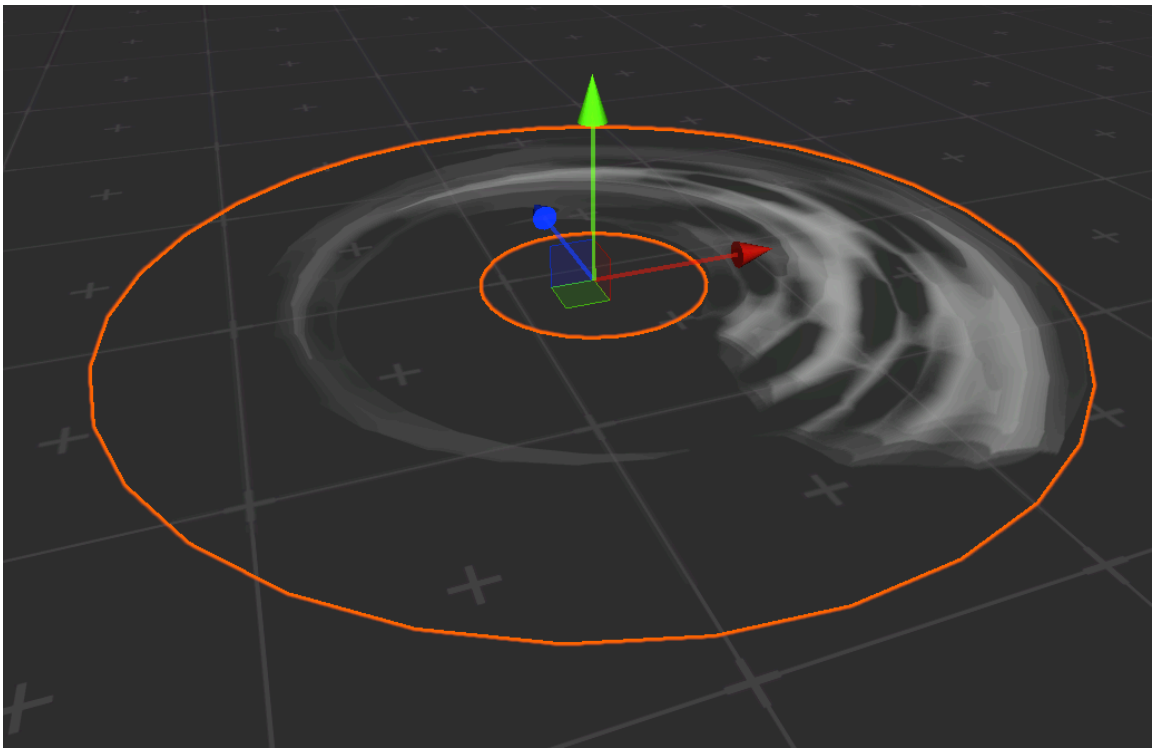
▼ Example



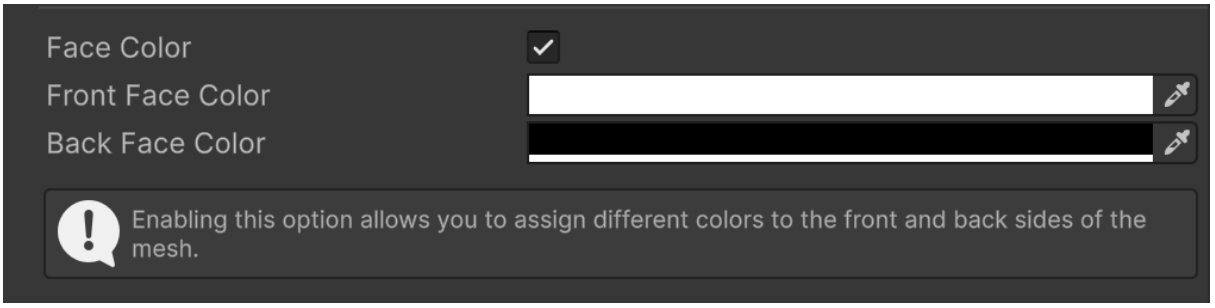


Step Mode





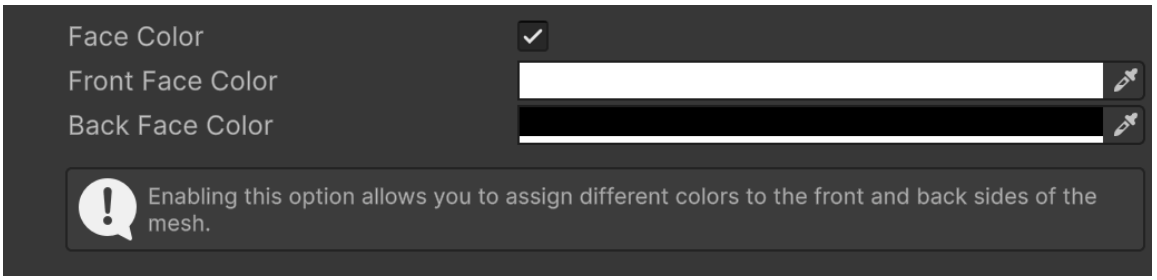
SmoothStep Mode

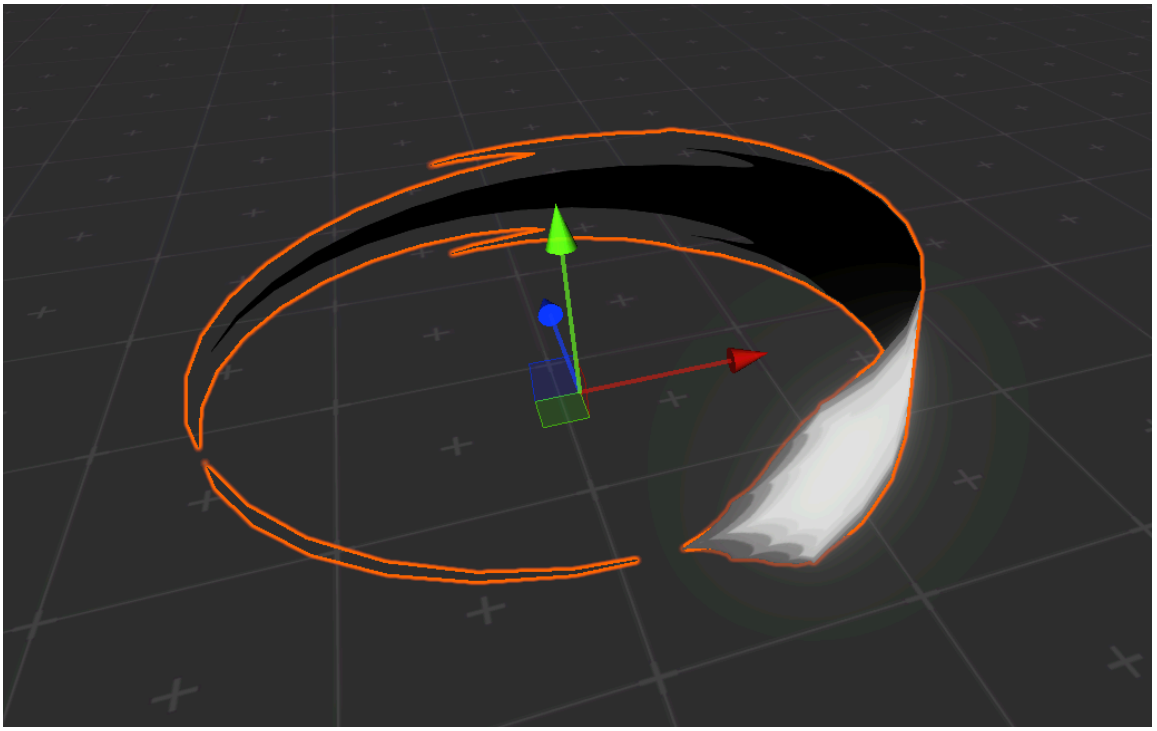


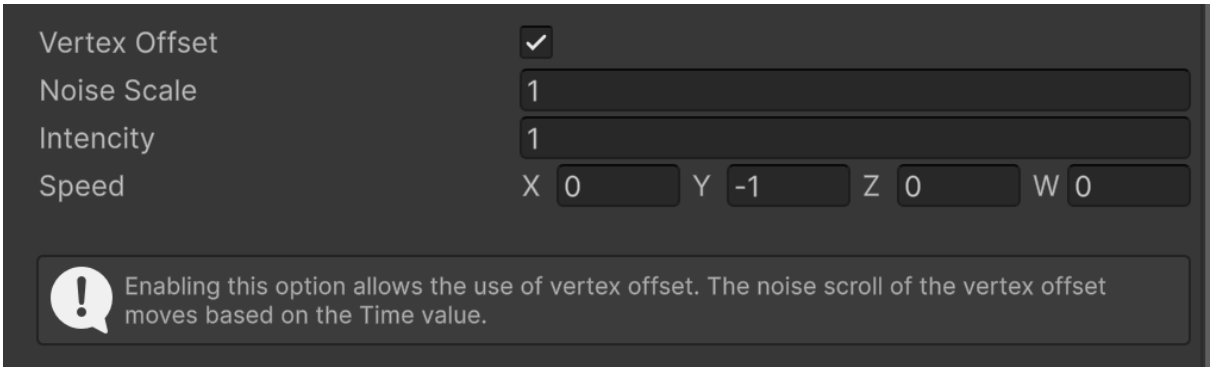
Enabling the Face Color toggle reveals settings related to face coloring.

Front Face Color	This sets the color applied to surfaces facing the normal direction.
Back Face Color	This sets the color applied to surfaces facing opposite the normal direction.

▼ Example

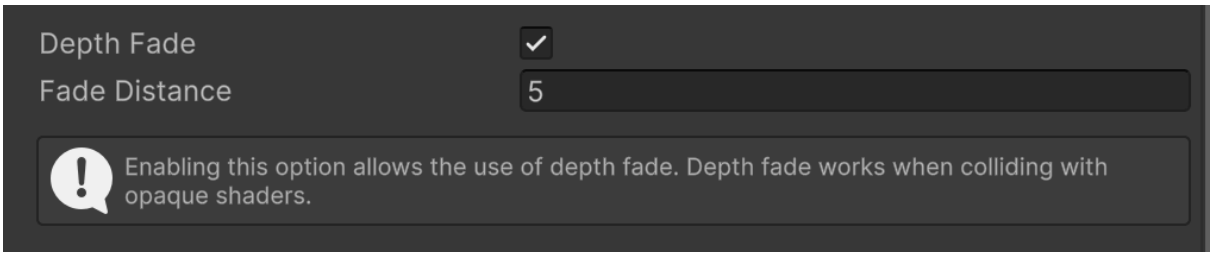






Enabling the Vertex Offset toggle reveals settings related to vertex offset. Vertex offset is driven by noise and animated based on the Time value.

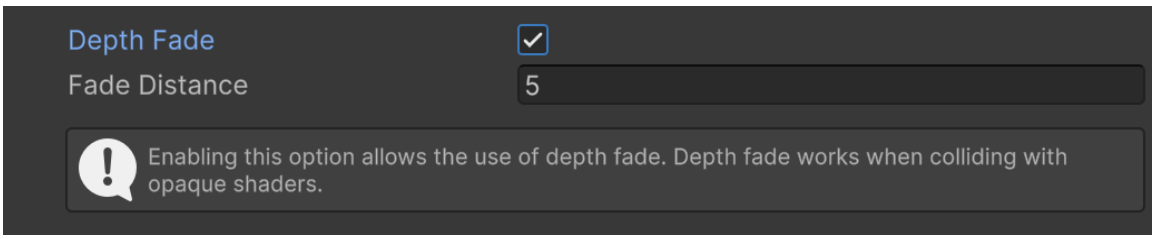
Noise Scale	Adjusts the scale of the noise pattern.
Intencity	Sets the strength of the vertex offset.
Speed	Controls the direction and speed of noise movement using the Speed value.

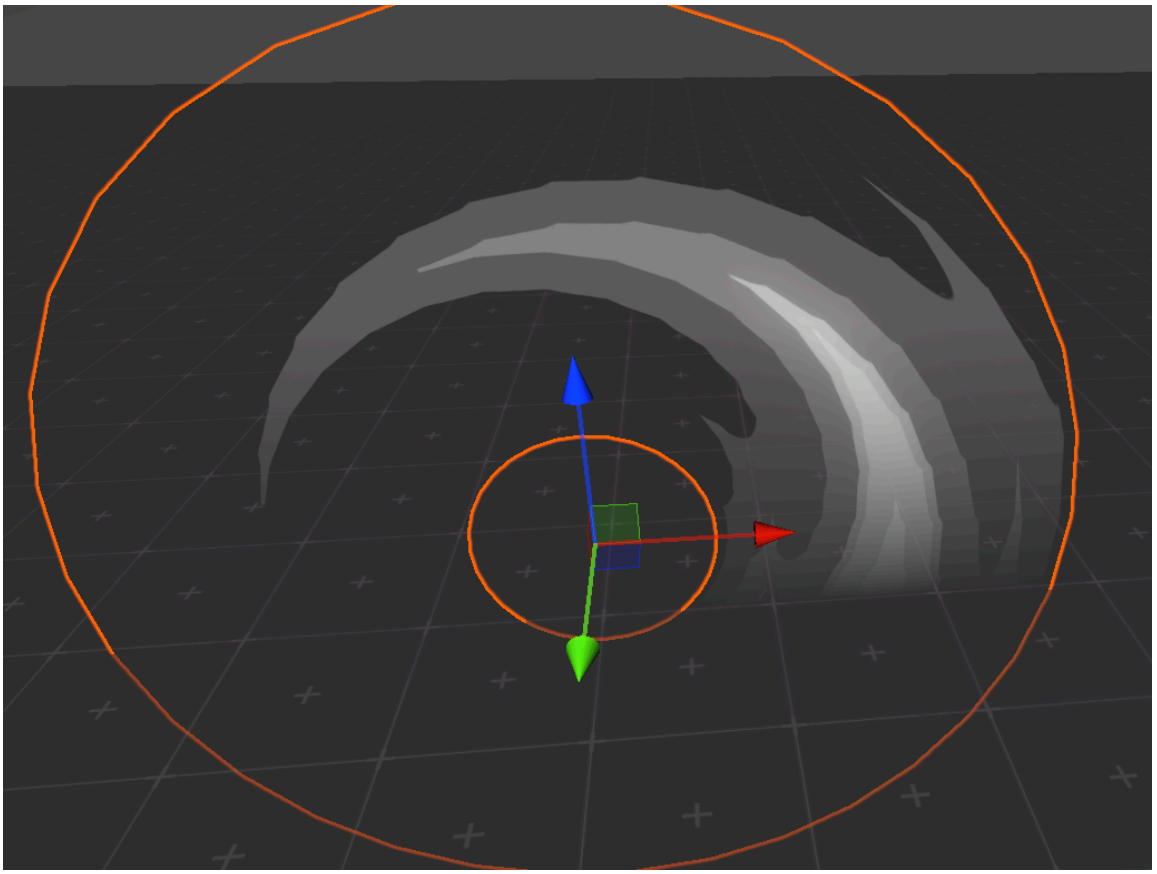


Enabling the Depth Fade toggle reveals settings related to depth fade. Depth Fade is applied when transparent and opaque surface types intersect.
If this condition is not met, the effect will not be applied.

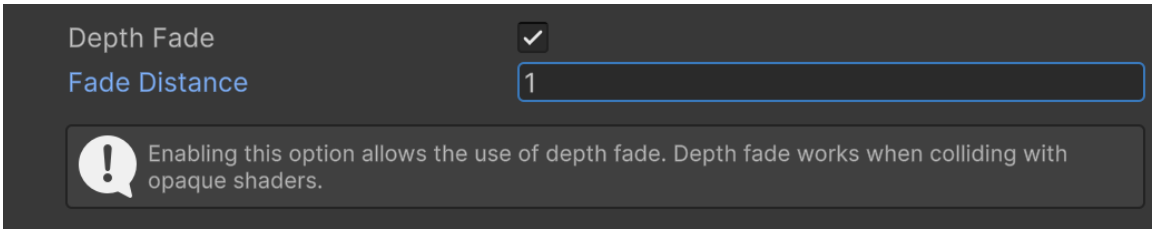
Fade Distance	Higher values result in a wider and smoother fade area.

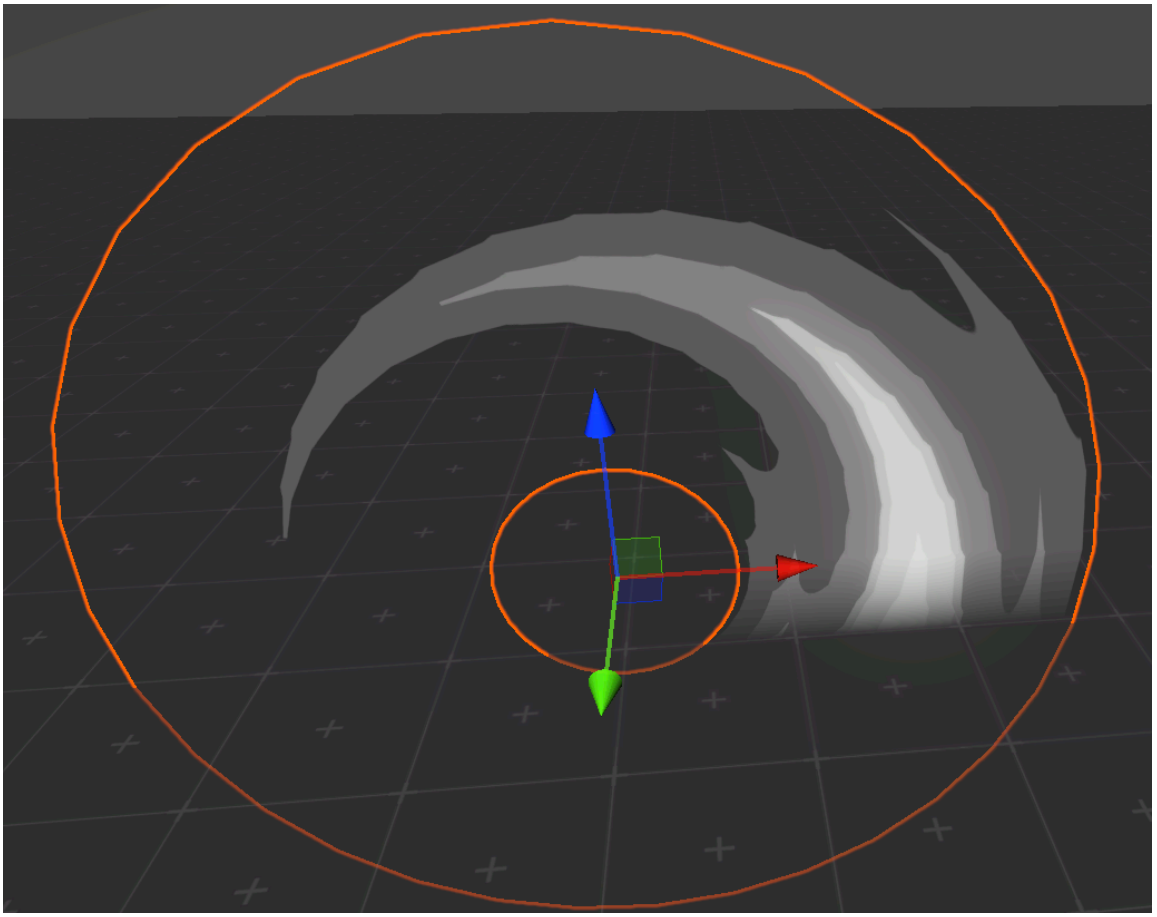
▼ Example



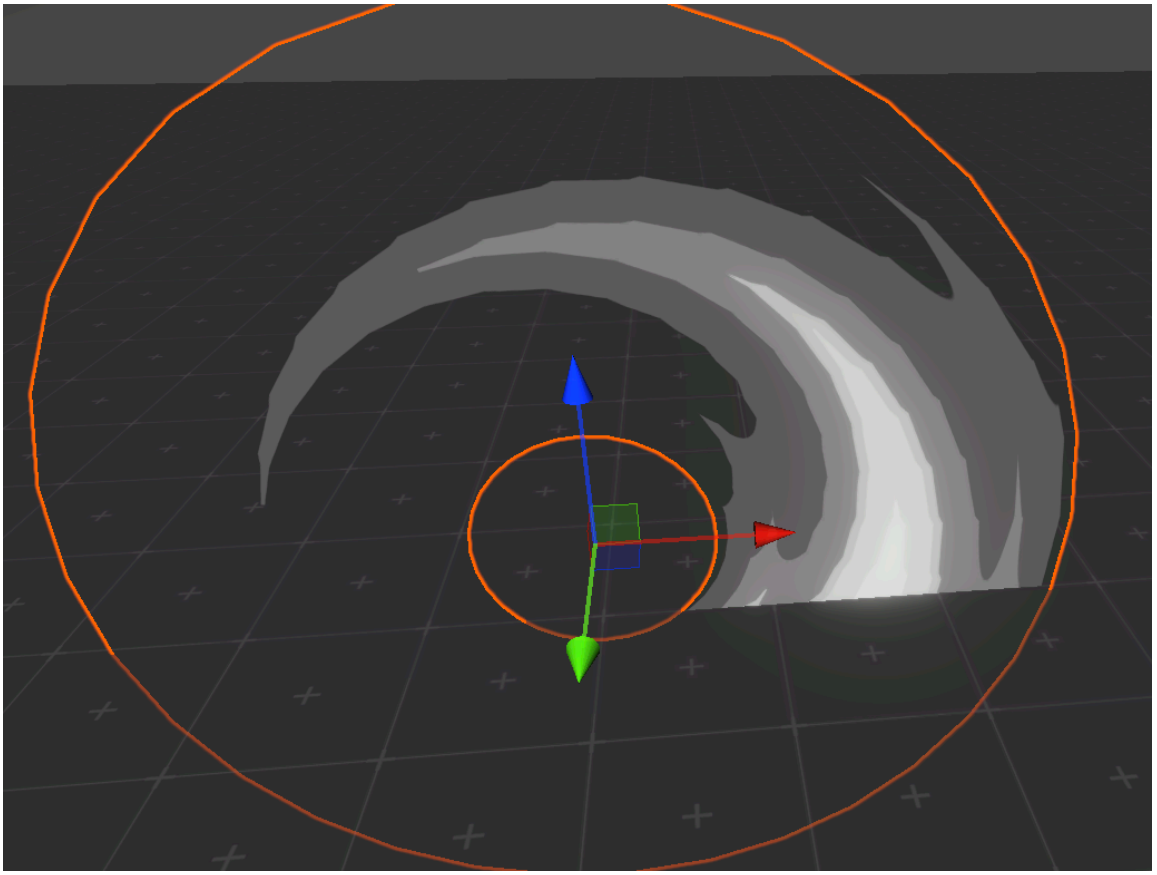


Fade Distance 5

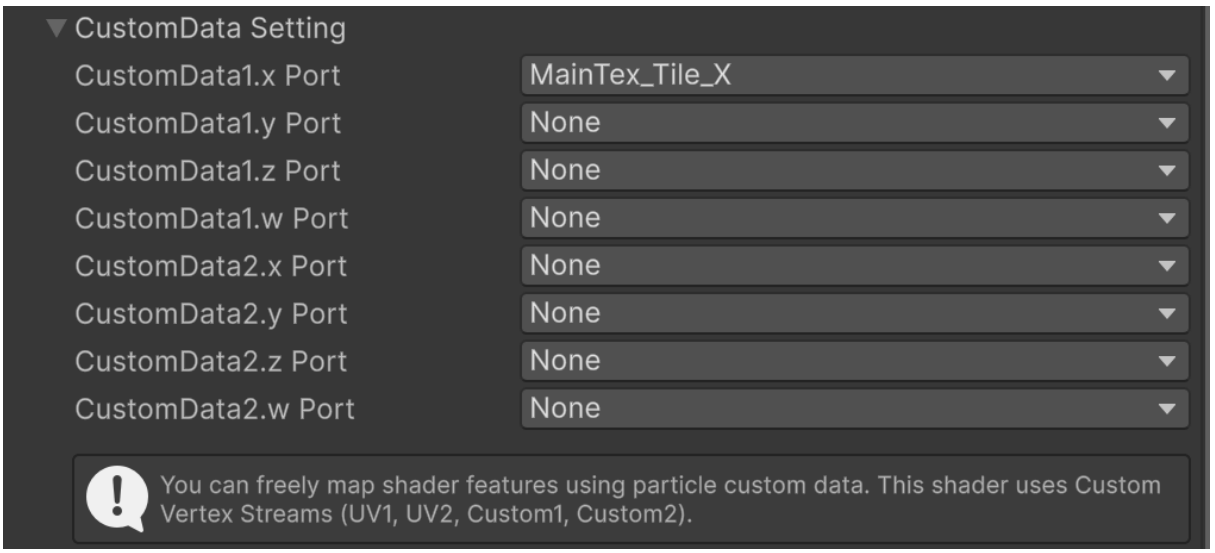


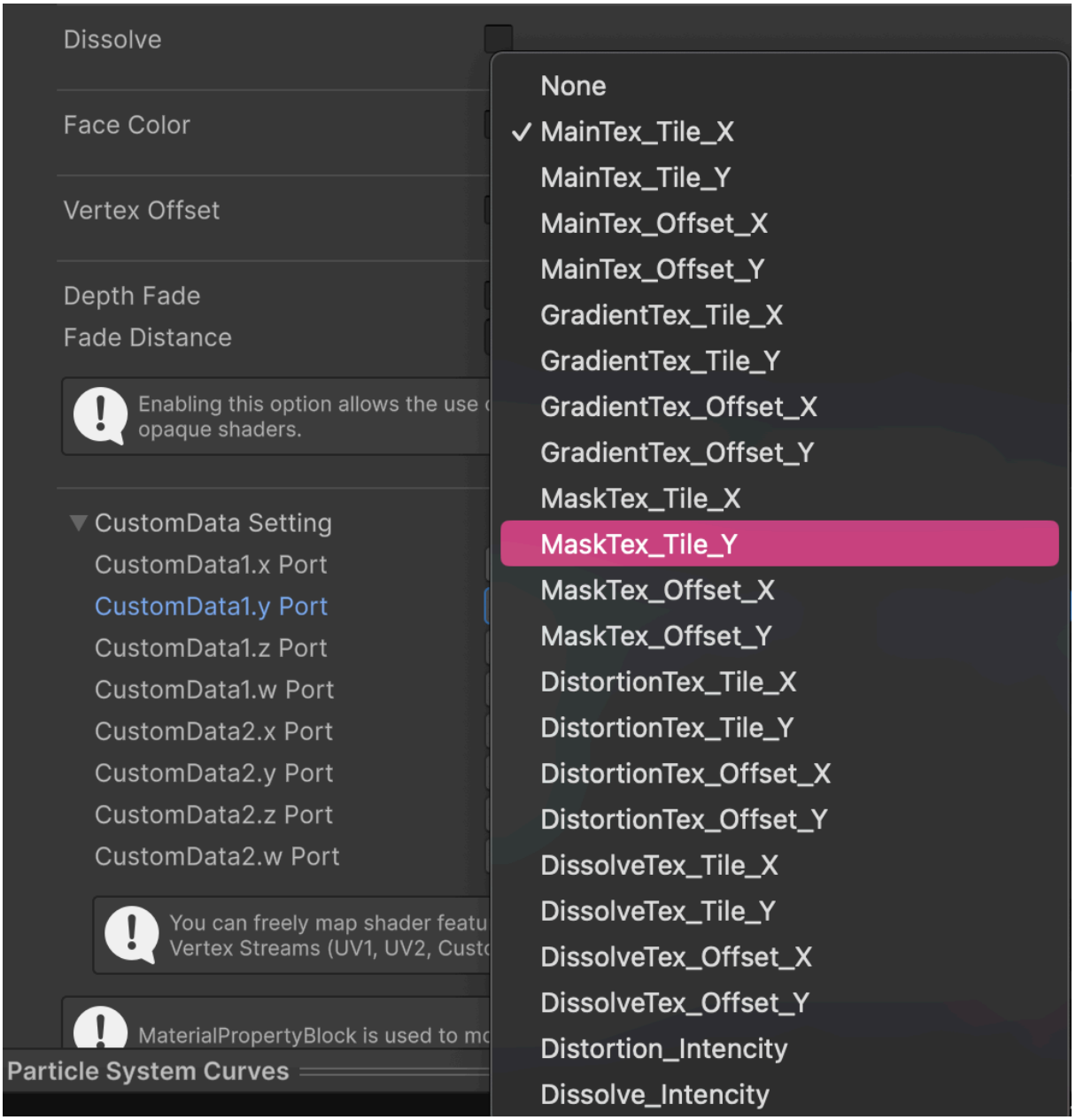


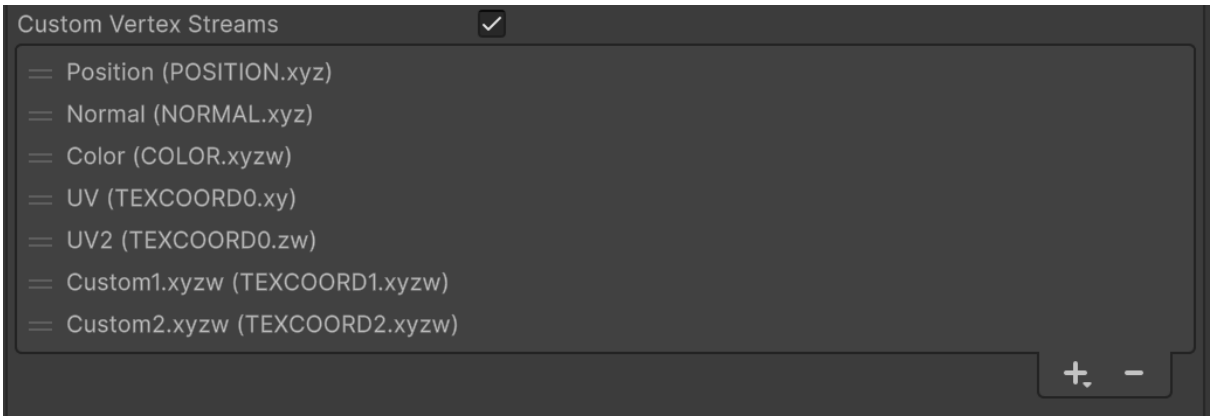
Fade Distance 1

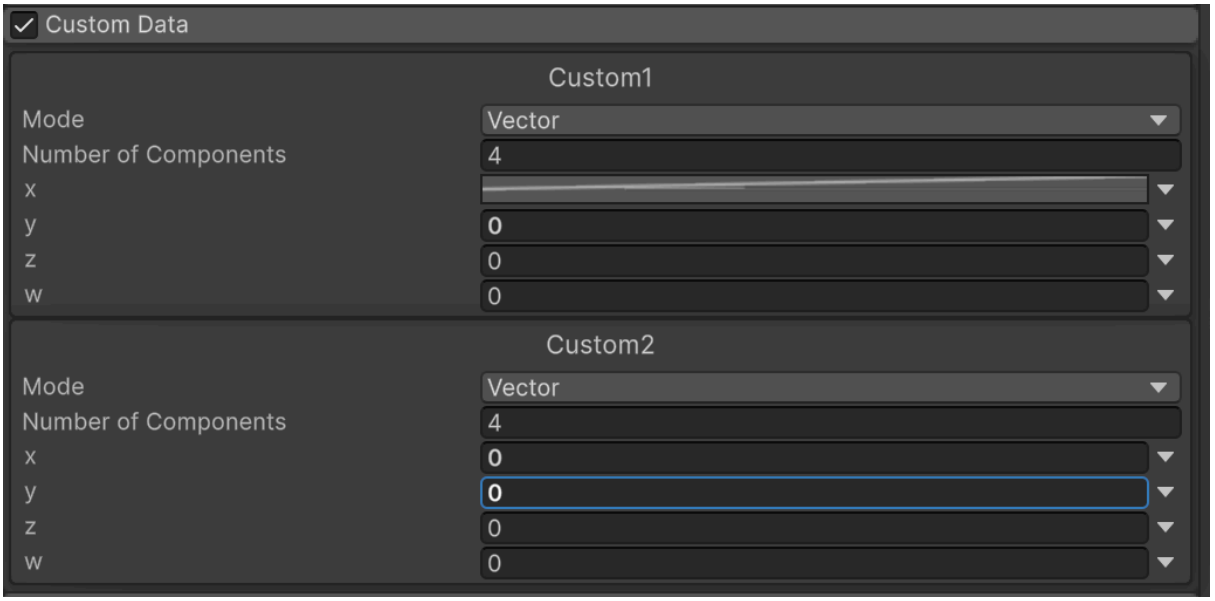


Disable Depth Fade




Dissolve
Face Color
Vertex Offset
Depth Fade
Fade Distance
Enabling this option allows the use
opaque shaders.
CustomData Setting
CustomData1.x Port
CustomData1.y Port
CustomData1.z Port
CustomData1.w Port
CustomData2.x Port
CustomData2.y Port
CustomData2.z Port
CustomData2.w Port
You can freely map shader featu
Vertex Streams (UV1, UV2, Custo
MaterialPropertyBlock is used to mo
Particle System Curves
None
MainTex_Tile_X
MainTex_Tile_Y
MainTex_Offset_X
MainTex_Offset_Y
GradientTex_Tile_X
GradientTex_Tile_Y
GradientTex_Offset_X
GradientTex_Offset_Y
MaskTex_Tile_X
MaskTex_Tile_Y
MaskTex_Offset_X
MaskTex_Offset_Y
DistortionTex_Tile_X
DistortionTex_Tile_Y
DistortionTex_Offset_X
DistortionTex_Offset_Y
DissolveTex_Tile_X
DissolveTex_Tile_Y
DissolveTex_Offset_X
DissolveTex_Offset_Y
Distortion_Intencity
Dissolve_Intencity


Custom Vertex Streams
Position (POSITION.xyz)
Normal (NORMAL.xyz)
Color (COLOR.xyzw)
UV (TEXCOORD0.xy)
UV2 (TEXCOORD0.zw)
Custom1.xyzw (TEXCOORD1.xyzw)
Custom2.xyzw (TEXCOORD2.xyzw)



Shader features can be freely arranged and utilized via particle custom data in the splash shader.
Since the shader uses particle custom vertex streams, make sure the Custom Vertex Streams in the Renderer Module are configured as shown in the reference image.
Once properly set, shader features can be controlled via the Custom Data Module. Please refer to the Shader Feature example included in the demo.